

Workflow Management Evolution for the HL-LHC Era

Traditional WLCG Model

- CPU-centric slot scheduling
- Static pledged resources
- High-throughput batch workflows
- Homogeneous worker nodes

Emerging Landscape

- Increasing orchestration complexity
- Hybrid resource models (allocation, cloud)
- Emerging workflow paradigms (AI related)
- Heterogeneous resources (GPU, ARM)

Takeaway: Workflow Management systems must evolve toward capability-aware orchestration of dynamic and heterogeneous resource pools.

Strategic Areas for Coordinated Evolution

Capability-Aware Scheduling

- Hardware-aware matchmaking
- Hybrid CPU/GPU workflows
- Whole-node and accelerator-aware execution

Fluid Resource Integration

- HPC and cloud allocations
- Opportunistic capacity
- Dynamic workload buffering

Emerging Workflow Models

- ML training and inference
- Interactive analysis facilities
- Flexible workflow orchestration

Cross-Experiment Coordination

- Shared provisioning concepts
- Common resource catalogues
- Coordination and standardisation

Bottom Line: Coordinated evolution across experiments and infrastructure providers will be essential to support HL-LHC computing requirements.